Experiment 12

Security and Governance in the Transport Management System

Aim

To implement security and governance in the Transport Management System using role-based access control, user permissions, data protection and workflow restrictions so that vehicle, driver and passenger information is accessed only by authorized users.

Procedure

1. **Open the application:** log in to Zoho Creator and open the Transport Management System.
2. **Create user roles:** define roles such as Transport Manager, Transport Coordinator, Driver and Route Planner.
3. **Configure permissions:** assign suitable permissions to each role for viewing, adding, editing and deleting records.
4. **Restrict data access:** configure report permissions and record-level access rules to protect driver mobile numbers, licence details and passenger contact details.
5. **Secure forms:** make essential fields mandatory and validate inputs to prevent incomplete or invalid submissions.
6. **Configure authentication:** use the available authentication settings so only authorized users can access the application.
7. **Protect workflows:** review Deluge scripts, email notifications and integrations to prevent unauthorized data access or unintended actions.
8. **Audit application activity:** use audit logs and monitoring features to review important changes and investigate suspicious activity.
9. **Test security:** log in with different roles and verify each user can access only the features and records permitted.

Access Matrix

Role	Access	View	Edit	Delete	Export
Transport Manager	All	Yes	Yes	Yes	Yes
Transport Coordinator	Forms, reports, page	Yes	Yes	No	No
Driver	Kanban, assigned trips	Own	Status only	No	No
Route Planner	Route form, map report	Limited	Routes only	No	No
Passenger (Portal)	Own trip details	Own	No	No	No

Security Test Cases

Test	Expected Result
Passenger opens another passenger's record	Not visible; only own records are listed
Driver tries to delete a trip	Action is not available
Submit form without mandatory fields	Blocked with validation message
Review the Metrics page	User access and activity are listed for auditing

Process Diagram

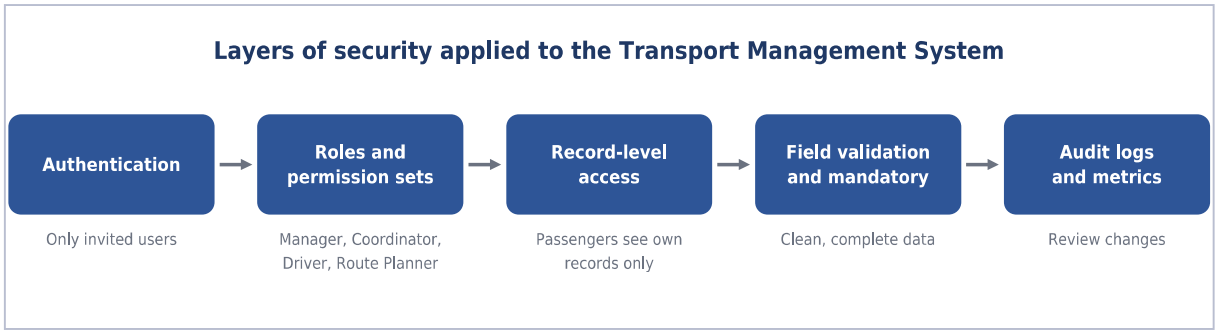


Figure 12.1: Diagram of the security layers (illustrative diagram, not a screenshot).

Permission Matrix by Role

Module	Transport Manager	Transport Coordinator	Driver	Route Planner	Passenger (Portal)
Trip_Scheduling form	Add, View, Edit, Delete	Add, View, Edit	View	No access	View own
Trip report	All records	All records	Assigned trips	Route fields only	Own records
Dashboard page	Yes	Yes	No	No	No
Export to Excel	Yes	No	No	No	No

Governance Practices

- Give each user only the access their job needs.
- Review audit logs and user activity regularly.
- Keep email notifications and integrations free of sensitive data such as licence and mobile numbers.
- Deactivate users who leave the organisation.

Result

Security and governance measures for the Transport Management System were defined through user roles, access permissions, data validation and activity monitoring.